ECE 3210 hw01 [SOLUTIONS]

1. Consider the following signals defined using unit step functions.

   (a) Sketch the following function

$$f_1(t) = t^2\big[u(t-1) - u(t-2)\big].$$

Solution:

$$f_1(t) = \begin{cases} t^2 & 1 \leq t < 2 \\ 0 & \text{otherwise} \end{cases}$$

   (b) Sketch the following function

$$f_2(t) = (t-4)\big[u(t-2) - u(t-4)\big].$$

Solution:

$$f_2(t) = \begin{cases} t-4 & 2 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

The signal is zero at $t = 4$ and equals -2 at $t = 2$.

2. Evaluate the following integrals. Some answers will be a numerical value, while others will be a function of t.

   (a)
$$\int_{-\infty}^{\infty} \delta(t)\, e^{-j\omega t}\, dt$$

   Solution: By the sifting property, evaluating $e^{-j\omega t}$ at $t = 0$: 1.

   (b)
$$\int_{-\infty}^{\infty} \delta(2t-3)\, \sin(\pi t)\, dt$$

   Solution: Scale: $\delta(2t-3) = \frac{1}{2}\delta\big(t - \frac{3}{2}\big)$. Sifting at $t = \frac{3}{2}$: $\frac{1}{2}\sin\big(\frac{3\pi}{2}\big) = \frac{1}{2}(-1) = -\frac{1}{2}$.

   (c)
$$\int_{-\infty}^{\infty} \delta(t+3)\, e^{-t}\, dt$$

   Solution: Sifting at $t = -3$: $e^{-(-3)} = e^{3}$.

   (d)
$$\int_{-\infty}^{\infty} (t^3+4)\, \delta(1-t)\, dt$$

   Solution: $\delta(1-t) = \delta(t-1)$; sifting at $t = 1$: $(1)^3 + 4 = 5$.

   (e)
$$\int_{-\infty}^{\infty} \delta(\tau)\, \sin(t-\tau)\, d\tau$$

   Solution: Sifting at $\tau = 0$: $\sin(t-0) = \sin(t)$.

   (f)
$$\int_{-\infty}^{\infty} \sin(\tau)\, \delta(t-\tau)\, d\tau$$

Solution: Sifting at $\tau = t$: $\sin(t)$. (Convolution of $\sin(t)$ with $\delta(t)$ reproduces the original.)

3. Consider the signal

$$x(t) = \begin{cases} t & 0 \le t < 1 \\ 0.5 + 0.5\cos(2\pi t) & 1 \le t < 2 \\ 3 - t & 2 \le t < 3 \\ 0 & \text{otherwise} \end{cases}$$

The energy of $x(t)$ is approximately $E_x \approx 1.0417$.

(a) What is the energy of $y_1(t) = \frac{1}{3}\, x(2t)$?

Solution: Going back to the integral definition of energy, we have

$$\begin{aligned} E_{y_1} &= \int_{-\infty}^{\infty} |y_1(t)|^2\, dt \\ &= \int_{-\infty}^{\infty} \left| \frac{1}{3}\, x(2t) \right|^2\, dt \end{aligned}$$

and

$$E_{y_1} = \int_{-\infty}^{\infty} \frac{1}{9}\, |x(2t)|^2\, dt$$

Let $u = 2t \implies du = 2\, dt \implies dt = \frac{1}{2}\, du$ with the limits of integration not changing. Then

$$\begin{aligned} E_{y_1} &= \int_{-\infty}^{\infty} \frac{1}{9}\, x(u)\, \frac{1}{2}\, du \\ &= \frac{1}{18} \int_{-\infty}^{\infty} x(u)\, du. \end{aligned}$$

This is the same as $\frac{1}{18} E_x$, and since $E_x \approx 1.0417$, we have

$$E_{y_1} = \frac{1}{18}\, E_x \approx \frac{1.0417}{18} \approx 0.057872.$$

(b) A periodic signal $y_2(t)$ is defined with period $T = 4$ s as

$$y_2(t) = \begin{cases} x(t) & 0 \le t < 4 \\ y_2(t+4) & \forall\, t \end{cases}.$$

What is the *power* of $y_2(t)$?

Solution: Since $x(t) = 0$ for $3 \leq t < 4$, one full period of y_2 has energy $E_x \approx 1.0417$. Power equals energy per period:

$$P_{y_2} = \frac{E_x}{T} = \frac{1.0417}{4} \approx 0.2604$$

(c) What is the *power* of $y_3(t) = \frac{1}{3}\, y_2(2t)$?

Solution: $y_2(2t)$ has the same power as $y_2(t)$ (time scaling preserves power of periodic signals). Amplitude scaling by $\frac{1}{3}$ gives:

$$P_{y_3} = \frac{1}{9}\, P_{y_2} = \frac{0.2604}{9} \approx 0.0289$$

4. For each system below with input $x(t)$ and output $y(t)$, determine whether the system is **linear** or **nonlinear**.

   (a)
$$\frac{dy(t)}{dt} + 2y(t) = x^2(t)$$

Solution: Nonlinear. Let $x(t) = ax_1(t) + bx_2(t)$ and suppose the corresponding output were $y(t) = ay_1(t) + by_2(t)$, where

$$\frac{dy_i(t)}{dt} + 2y_i(t) = x_i^2(t), \quad i = 1, 2.$$

Then

$$\frac{d}{dt}\big(ay_1 + by_2\big) + 2\big(ay_1 + by_2\big) = ax_1^2 + bx_2^2.$$

But the system driven by $ax_1 + bx_2$ requires

$$\begin{aligned}\frac{dy}{dt} + 2y &= (ax_1 + bx_2)^2 \\ &= a^2x_1^2 + 2ab\,x_1x_2 + b^2x_2^2,\end{aligned}$$

which is generally not equal to $ax_1^2 + bx_2^2$. Therefore superposition fails, so the system is nonlinear.

   (b)
$$\frac{dy(t)}{dt} + 3t\,y(t) = t^2x(t)$$

Solution: Linear. Let $x(t) = ax_1(t) + bx_2(t)$ and let y_1, y_2 satisfy

$$\frac{dy_i}{dt} + 3t\,y_i = t^2x_i, \quad i = 1, 2.$$

Define $y = ay_1 + by_2$. Then

$$\begin{aligned}\frac{dy}{dt} + 3t\,y &= a\left(\frac{dy_1}{dt} + 3t\,y_1\right) + b\left(\frac{dy_2}{dt} + 3t\,y_2\right) \\ &= at^2x_1 + bt^2x_2 \\ &= t^2(ax_1 + bx_2) \\ &= t^2x.\end{aligned}$$

So the output for input $ax_1 + bx_2$ is $ay_1 + by_2$, hence superposition holds and the system is linear.

(c)

$$5y(t) + 1 = x(t)$$

Solution: Nonlinear. Let $x_i \mapsto y_i$ satisfy

$$5y_i + 1 = x_i, \quad i = 1, 2.$$

For $x = ax_1 + bx_2$, if superposition were true we would have $y = ay_1 + by_2$. Then

$$\begin{aligned} 5y + 1 &= 5(ay_1 + by_2) + 1 \\ &= a(5y_1) + b(5y_2) + 1 \\ &= a(x_1 - 1) + b(x_2 - 1) + 1 \\ &= ax_1 + bx_2 - (a + b) + 1. \end{aligned}$$

For this to equal $x = ax_1 + bx_2$ for all a, b, we would need $-(a + b) + 1 = 0$ always, which is false. Therefore superposition fails, so the system is nonlinear.

(d)

$$\left(\frac{dy(t)}{dt}\right)^2 + 2y(t) = x(t)$$

Solution: Nonlinear. Let $x_i \mapsto y_i$ satisfy

$$\left(\frac{dy_i}{dt}\right)^2 + 2y_i = x_i, \quad i = 1, 2.$$

For input $x = ax_1 + bx_2$, test $y = ay_1 + by_2$:

$$\left(\frac{d}{dt}(ay_1 + by_2)\right)^2 + 2(ay_1 + by_2) = \left(a\frac{dy_1}{dt} + b\frac{dy_2}{dt}\right)^2 + 2ay_1 + 2by_2.$$

Expanding the square gives

$$a^2\left(\frac{dy_1}{dt}\right)^2 + 2ab\frac{dy_1}{dt}\frac{dy_2}{dt} + b^2\left(\frac{dy_2}{dt}\right)^2 + 2ay_1 + 2by_2,$$

which is generally not equal to

$$\begin{aligned} &a\left[\left(\frac{dy_1}{dt}\right)^2 + 2y_1\right] + b\left[\left(\frac{dy_2}{dt}\right)^2 + 2y_2\right] \\ &= ax_1 + bx_2. \end{aligned}$$

Hence superposition fails and the system is nonlinear.

(e)

$$\frac{dy(t)}{dt} + \sin(t)\, y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Solution: Linear. Let $x(t) = ax_1(t) + bx_2(t)$, and let y_1, y_2 satisfy

$$\frac{dy_i}{dt} + \sin(t)\, y_i = \frac{dx_i}{dt} + 2x_i, \quad i = 1, 2.$$

Set $y = ay_1 + by_2$. Then

$$\begin{aligned}
\frac{dy}{dt} + \sin(t)\, y &= a\left(\frac{dy_1}{dt} + \sin(t)\, y_1\right) + b\left(\frac{dy_2}{dt} + \sin(t)\, y_2\right) \\
&= a\left(\frac{dx_1}{dt} + 2x_1\right) + b\left(\frac{dx_2}{dt} + 2x_2\right) \\
&= \frac{d}{dt}(ax_1 + bx_2) + 2(ax_1 + bx_2) \\
&= \frac{dx}{dt} + 2x.
\end{aligned}$$

So $x = ax_1 + bx_2$ produces $y = ay_1 + by_2$. Superposition holds, so the system is linear.

(f)

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Solution: Linear. Let $x(t) = ax_1(t) + bx_2(t)$ and let y_1, y_2 satisfy

$$\frac{d^2y_i}{dt^2} + 5\frac{dy_i}{dt} + y_i = \frac{dx_i}{dt} + 2x_i, \quad i = 1, 2.$$

Define $y = ay_1 + by_2$. Then

$$\begin{aligned}
\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + y &= a\left(\frac{d^2y_1}{dt^2} + 5\frac{dy_1}{dt} + y_1\right) + b\left(\frac{d^2y_2}{dt^2} + 5\frac{dy_2}{dt} + y_2\right) \\
&= a\left(\frac{dx_1}{dt} + 2x_1\right) + b\left(\frac{dx_2}{dt} + 2x_2\right) \\
&= \frac{d}{dt}(ax_1 + bx_2) + 2(ax_1 + bx_2) \\
&= \frac{dx}{dt} + 2x.
\end{aligned}$$

Therefore the output for $ax_1 + bx_2$ is $ay_1 + by_2$, so superposition holds and the system is linear.

5. For each system below with input $x(t)$ and output $y(t)$, determine whether the system is **time-invariant (TI)** or **not time-invariant (not TI)**.

(a)

$$y(t) = x(t-2)$$

Solution: Time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t-t_0)$. Then

$$\begin{aligned}\tilde{y}(t) &= \tilde{x}(t-2)\\ &= x(t-t_0-2)\\ y(t-t_0) &= x((t-t_0)-2)\\ &= x(t-t_0-2).\end{aligned}$$

Since $\tilde{y}(t) = y(t-t_0)$, the system is time-invariant.

(b)

$$y(t) = \log\big(x(t)\big)$$

Solution: Time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t-t_0)$. Then

$$\begin{aligned}\tilde{y}(t) &= \log(x(t-t_0))\\ y(t-t_0) &= \log(x(t-t_0)).\end{aligned}$$

Since $\tilde{y}(t) = y(t-t_0)$, the system is time-invariant.

(c)

$$y(t) = x(t)^2$$

Solution: Time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t-t_0)$. Then

$$\begin{aligned}\tilde{y}(t) &= x(t-t_0)^2\\ y(t-t_0) &= x(t-t_0)^2.\end{aligned}$$

Since $\tilde{y}(t) = y(t-t_0)$, the system is time-invariant.

(d)

$$y(t) = \int_{-5}^{5} x(\tau)\, d\tau$$

Solution: Not time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t - t_0)$. Then

$$\tilde{y}(t) = \int_{-5}^{5} x(\tau - t_0)\, d\tau$$
$$y(t - t_0) = \int_{-5}^{5} x(\tau)\, d\tau \quad \text{(independent of } t\text{)}.$$

Since $\tilde{y}(t) \neq y(t - t_0)$ in general, the system is not time-invariant.

(e)

$$y(t) = t\, x(t)$$

Solution: Not time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t - t_0)$. Then

$$\tilde{y}(t) = t\, x(t - t_0)$$
$$y(t - t_0) = (t - t_0)\, x(t - t_0).$$

Since $\tilde{y}(t) \neq y(t - t_0)$, the system is not time-invariant.

(f)

$$y(t) = \cos(t)\, x(t) + a$$

Solution: Not time-invariant. Shift the input by t_0: $\tilde{x}(t) = x(t - t_0)$. Then

$$\tilde{y}(t) = \cos(t)\, x(t - t_0) + a$$
$$y(t - t_0) = \cos(t - t_0)\, x(t - t_0) + a.$$

Since $\tilde{y}(t) \neq y(t - t_0)$ in general, the system is not time-invariant.

6. A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)\, y(t) = (2D^3 + 1)\, x(t).$$

(a) Find the characteristic equation $Q(\lambda) = 0$ of the system.

Solution: Substitute λ for D in the left-hand side operator: $Q(\lambda) = \lambda^3 + 9\lambda$. Setting $Q(\lambda) = 0$: $\lambda(\lambda^2 + 9) = 0$, so $\lambda_1 = 0$, $\lambda_{2,3} = \pm 3j$. MOdes: 1, e^{3jt}, e^{-3jt}.

(b) Given the initial conditions $y_{\text{zir}}(0) = 4$, $y'_{\text{zir}}(0) = -18$, and $y''_{\text{zir}}(0) = 0$, determine the zero-input response $y_{\text{zir}}(t)$ for $t \geq 0$.

Solution: The general zero-input response is:

$$\begin{aligned} y_{\text{zir}}(t) &= c_1 + c_2 e^{-3jt} + c_3 e^{3jt} \\ &= c_1 + c_2(\cos(3t) - j\sin(3t)) + c_3(\cos(3t) + j\sin(3t)) \\ &= c_1 + (c_2 + c_3)\cos(3t) + j(c_3 - c_2)\sin(3t). \end{aligned}$$

Combining the constants, we have

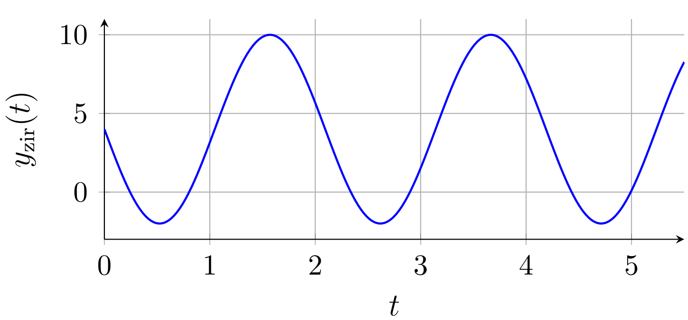
$$y_{\text{zir}}(t) = c_1 + c_4 \cos(3t) + c_5 \sin(3t), \quad t \geq 0.$$

Applying initial conditions:

$$\begin{aligned} y_{\text{zir}}(0) &= c_1 + c_4 = 4 \\ y'_{\text{zir}}(0) &= -3c_4 \sin(0) + 3c_5 \cos(0) = 3c_5 = -18 \implies c_5 = -6 \\ y''_{\text{zir}}(0) &= -9c_4 \cos(0) - 9c_5 \sin(0) = -9c_4 = 0 \implies c_4 = 0 \end{aligned}$$

From the first equation: $c_1 = 4$. Therefore

$$y_{\text{zir}}(t) = 4 - 6\sin(3t), \quad t \geq 0.$$



7. A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^2 + 4D + 4)\, y(t) = D\, x(t).$$

(a) Find the characteristic equation $Q(\lambda) = 0$ of the system.

> **Solution:** Substitute λ for D in the left-hand side: $Q(\lambda) = \lambda^2{+}4\lambda{+}4 = (\lambda{+}2)^2$.

(b) Find the characteristic roots (eigenvalues).

> **Solution:** $(\lambda + 2)^2 = 0 \implies \lambda_{1,2} = -2$ (repeated). Modes: $\{e^{-2t},\, t\,e^{-2t}\}$.

(c) Given the initial conditions $y_{\text{zir}}(0^-) = 3$ and $y'_{\text{zir}}(0^-) = -4$, determine the zero-input response $y_{\text{zir}}(t)$ for $t \geq 0$.

> **Solution:** Repeated root $\Rightarrow$ general ZIR: $y_{\text{zir}}(t) = (c_1 + c_2 t)\, e^{-2t}$. Applying initial conditions:
>
> $$\begin{aligned} y_{\text{zir}}(0^-) &= c_1 = 3 \\ y'_{\text{zir}}(0^-) &= c_2 - 2c_1 = -4 \implies c_2 = 2 \end{aligned}$$
>
> Therefore: $y_{\text{zir}}(t) = (3 + 2t)\, e^{-2t}, \quad t \geq 0.$